SLIDE 1

Problem & why it matters

TEAM Knock Before Noon

PROBLEM STATEMENT B1 — HeatGuard: Urban heat risk mapping for vulnerable communities

THE PROBLEM IN ONE SENTENCE

Singapore's heat is getting more dangerous for the seniors least able to cope, but today's advisories say how hot it will be, not who needs a check-in, so outreach teams react during the heat instead of before it.

WHO IS AFFECTED, AND HOW BADLY

Seniors living alone: 87,200 one-person households aged 65+ in 2024, double the 42,100 of 2014.

Seniors in 1–2-room flats or with mobility limits: least able to cool down or reach help (Census 2020 tables).

Eldercare outreach teams: no early warning of which areas to prioritise.

WHY IT MATTERS NOW

2025 was Singapore's hottest year on record; **heat-illness ED cases rose 42%** (280 → 398).

2026 is the **Year of Climate Adaptation** and the year Singapore turns super-aged; a S$40m heat-resilience programme names the elderly a priority.

87,200

seniors aged 65+ living alone (2024)

2× in ten years

+42%

heat-illness ED cases, 2024 → 2025

280 → 398 cases

1 in 5

residents aged 65+ in 2026

Singapore becomes super-aged

Sources: MOH parliamentary reply (Jan 2026), seniors living alone; MOH via The Straits Times (2026), heat-illness ED cases; MSE Year of Climate Adaptation 2026; DAISI problem brief B1.

Solution & data

OUR SOLUTION IN ONE SENTENCE

Every afternoon, HeatGuard forecasts tomorrow's heat stress for each of Singapore's ~330 subzones, combines it with who lives there, and tells outreach teams where to check on seniors first, and why.

WHO USES IT, AND WHAT DECISION IT CHANGES

- **Active Ageing Centre coordinators** → which clients to call first, before tomorrow's peak heat
- **Silver Generation Office ambassadors** → which neighbourhoods get door-knock rounds
- **Community Clubs (People's Association)** → where to open as a cool rest spot, and when

HeatGuard says where and when; each team's own client list says who. No personal data is used.

DATASETS WE WILL USE (all on data.gov.sg)

	Dataset	Question it answers
1	**Heat stress (WBGT) observations** NEA · live API · 30 stations, every 15 min, since Feb 2025	*Heat now + what we predict*
2	**Air temperature & relative humidity** NEA · live API + yearly files 2016–2024	*More hot seasons to learn from*
3	**24-hour forecast & 4-day outlook** NEA · live API + archives	*What NEA expects tomorrow*
4	**Census 2020: seniors 65+, flat type, living alone, mobility** SingStat · by subzone or planning area	*Who lives there? How vulnerable?*
5	**Subzone map + Community Clubs & public libraries** URA · People's Association · NLB	*Nearest air-conditioned space?*

DIFFERENT FROM A GENERIC DASHBOARD

An action list, not just a map
Ranked subzones, each with a reason and a suggested next step.

Tomorrow, not today
Next-day heat-stress forecast, scored vs. a "same as today" baseline.

Honest about the data
We checked every dataset (e.g. 16 temperature stations, not ~50) and show which inputs are estimates.

Tomorrow's check-in list *illustrative example, not real output*

#	Subzone	2–4 pm	Why	Suggested action
1	Bedok North	HIGH	Many seniors 65+ · many live alone	Call clients who live alone by 11 am
2	Toa Payoh Central	HIGH	Many 1–2-room rental flats	Door-knock rental blocks; share CC hours
3	Jurong West Central	MODERATE	Nearest air-con space over 1 km	Offer escort to nearest CC or library

Rank = tomorrow's heat band × how vulnerable the area is. Actions come from a fixed rulebook, not AI.

Risk bands reuse NEA's own Low / Moderate / High heat-stress categories. All datasets are open data from data.gov.sg.

SLIDE 3

Databricks architecture & impact

PIPELINE: SOURCE → INGESTION → TRANSFORM → MODEL / ANALYTICS → OUTPUT

1 SOURCE

data.gov.sg

Live APIs: heat stress, temperature, humidity, NEA forecasts
Files: Census 2020, subzone map, CCs, libraries

2 INGEST

Lakeflow Jobs + Auto Loader

Job pulls live readings every 15 min into a **Unity Catalog Volume** → bronze Delta tables

3 TRANSFORM

Lakeflow Declarative Pipeline

Quality checks (missing, duplicate, impossible readings); **Spatial SQL + H3:** stations → subzones, distance to nearest CC / library

4 MODEL

MLflow

① Tomorrow's heat stress per station
② Vulnerability score per subzone
③ Priority = heat × vulnerability

5 OUTPUT

AI/BI + Genie

3-tab dashboard for coordinators; **Genie** answers plain-English questions; **Foundation Model API** drafts advisories

Unity Catalog governance: lineage from raw API to final list · area-level data only, no personal data · every source and known gap documented

THE DASHBOARD WE WILL DEMO

NOW	Live heat stress at 30 stations in NEA's Low / Moderate / High bands, refreshed every 15 min.
TOMORROW	Subzone risk map + the ranked check-in list, with a reason and a suggested action per area.
WHY HERE?	Click an area: what drove its rank, its nearest CC or library, and past forecast accuracy.

Plus: ask Genie "Which High subzones in Bedok are over 1 km from a CC?" and get advisory text in one click.

MEASURABLE IMPACT IF IT WORKED

~24 h lead time: outreach planned the day before, not during the heat.

Focus: the top ~20 of ~330 subzones where high heat meets vulnerable seniors.

Honest accuracy: forecast error vs. "same as today", tracked in MLflow.

Outcome to track: heat-illness ED cases (398 in 2025).

TWO-WEEK SPRINT PLAN

Days 1–2	Ingestion + history backfill
Days 3–5	Pipeline, subzone mapping, vulnerability score
Days 6–9	Forecast model + baseline in MLflow
Days 10–12	Dashboard, Genie, advisories, governance
Days 13–14	End-to-end test, backup recording, pitch

Built on Databricks Free Edition: one Lakeflow pipeline and one serverless SQL warehouse, within free-tier quotas.